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Researching teachers' time use: Complexity, challenges and a possible way forward

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Abstract
In the context of global concerns about teacher workload and the relationship between workload and attrition, understanding the nature, quantity and intensity of teachers' work is an essential first step in formulating robust solutions to this significant problem. Understanding teachers' work, however, is a complex undertaking, and prior attempts have largely been focused on the quantity rather than the intensity or quality of work required and undertaken. This article reports on a pilot study of the Teacher Time Use app, a bespoke tool developed by the research team to 'get inside' teachers' subjective experience of time through a focus on both workload and intensity. Our analysis shows that the app provides a simple, non-demanding way for teachers to record their work in a timely and efficient way. It also highlights the capacity of this approach to understand both the range and quantum of tasks that comprise teachers' work and consequently the nature and subjective experience of work intensification. We argue the need for a more nuanced empirical understanding of the layering and multi-tasking of teachers' work that characterises work intensity, and suggest that the Teachers' Time Use app provides an effective means for recording and representing the complex dimensions of teachers' work and time use.

Keywords Workload · Work intensification · Time poverty · Teachers' work · Subjective experience

Introduction

Concern about work pressure is a perennial issue in many professions. Over the last 20 years, research on medicine, law and dentistry has highlighted ongoing concern around stress and burnout associated with workloads and work intensification. The global COVID19 pandemic brought this into stark relief with 'the great resignation' as burnout, feelings of overwork, and despair at the relentlessness of work pressure and frustration with a lack of improvement meant that many professionals decided to leave

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their careers (Sheather & Slattery, 2021). It would be a mistake, however, to suggest that the great resignation was caused primarily by COVID19, as research over decades has highlighted ongoing dissatisfaction with professional work and its impact on job satisfaction, health and wellbeing. This is particularly evident in the teaching profession as concerns about the sustainability of work are causing many teachers and principals to leave their positions (Heffernan et al., 2022; Reid & Creed, 2021). Globally, teaching is no longer seen as an attractive career (Thompson, 2021). Research has identified problems with teacher stress (McIntyre et al., 2017), wellbeing (Collie, 2021) and burnout (Van Droogenbroeck et al., 2021). Complementing these problems are concerns about teacher attrition (Weldon, 2018). In Australia, such anxieties have initiated policy efforts aimed at recruitment and, to a lesser degree, retention of teachers (Australian Government Department of Education, 2022).

While intensification is usually understood as a sense of greater time pressure in attending to tasks (Fitzgerald et al., 2019; Green, 2021), slippage between workload and work intensification is common, and the terms are not always explicitly delineated (Creagh, 2023). Work intensification, or 'heavy hours', is more difficult to measure than workload because it is a subjective experience. Our argument is that it is also a crucial factor in understanding why teaching has become such an unsustainable profession for many at this point in time. As Santoro and Hazel (2022) have argued, workload and affective or qualitative dimensions of teachers' work interact in complex ways impacting the experience of work for the teaching profession. In order to 'get inside' this problem, an appreciation of the complexity of teachers' work and the time pressures wrought not only by the amount of work required (workload) but concomitantly, the demands of that work by virtue of its density and the stakes attached (intensification), is required.

With this at the forefront of our thinking, we designed a 'timetracker app' where teachers record their time use across randomly selected 30-min segments. Our aim was to explore the complexity of teachers' work, not just as a list of activities or quantum of work, but to uncover the intensity or intensification of teachers' work through recording their subjective experience of time. This paper focuses specifically on a pilot conducted in schools in Queensland, Australia amid Australia-wide concerns about retention and attrition of teachers (Productivity Commission, 2022, p. 208). This paper proceeds in three parts. First, we provide background to the study including previous research on workload and work intensification and problems associated with researching the intensity of teachers' work. We then outline the pilot of the timetracker app, asking teachers to record their time use in short, 30-min, randomly selected periods. Finally, we argue that this approach shows potential in more accurately representing teachers' work intensification.

Workload, work intensification and problems accounting for teachers' work

Our specific interest lies in teachers' workload and work intensification as a systemic, or structural, problem. Systems create the conditions in which teachers work and this must be the starting point for understanding teachers' work and its relation

to retention, attrition and career desirability (Thompson, 2021). International studies report high working hours, sometimes in excess of 50 h per week (Allen et al., 2021; Manuel et al., 2018; Sato et al., 2020) while also reporting that workload is increasing, often attributed to the intrusion of 'non-core' tasks, such as administrative requirements (Fitzgerald et al., 2019; Gavin et al., 2021; Lawrence et al., 2019).

While workload and work intensification are global issues, measuring teachers' work remains challenging. Workload is emphasised in the literature because it is easier to measure through estimations of hours of work teachers typically do in a week. Examples include the OECD's Teaching and Learning International Survey (TALIS) which then ranks systems on how much work teachers report doing in a given week (OECD, 2018, 2019). The data produced lend themselves to reporting of, for example, average numbers of hours worked per week during term, as in the case of McGrath-Champ et al. (2018); or, in the case of TALIS, hours worked during teachers' 'most recent calendar week' (OECD, 2018).While this approach is useful in providing the 'gist' of teachers' experience (see Brainerd & Reyna, 1993), concerns have been expressed about accuracy as retrospective self-reports tend to overestimate 'core' activities and underestimate 'peripheral' activities (te Braak et al, 2022; te Braak, 2022). This methodological challenge can invite policy solutions unlikely to improve teachers' work conditions. This demonstrates that workload is not a sufficient concept for understanding those problems associated with teacher attrition and burnout (Santoro & Hazel, 2022).

The subjective experience of teaching time

One of the challenges with researching teacher time use concerns the relationship between the quantity of time (as measured through seconds, minutes, hours and so on) and the quality of that same quantity of time. This difference may be usefully illustrated through the concepts of workload and work intensification. As noted above, workload concerns the quantum of time taken or devoted to a particular task or occupation. Workload is usually a measure of hours spent over a given period. For example, the question on the OECD's TALIS survey, asking teachers to report the hours they work for a particular week (OECD, 2020) is illustrative because it suggests that workload is (a) measurable and (b) invariant across contexts. An 'objective' understanding of workload is possible because the unit of measure does not change. This depends heavily on what the French philosopher Henri Bergson (1991) called 'clock time', that is the way that time is organised spatially into measurable intervals. Clock time, as a social phenomenon is central to governing and governance, from industrial relations legislation to the organisation of factories and institutions (Thompson, 1967). Bergson (1991, 1999) juxtaposed spatialised, clock time with subjective or 'lived time'. Lived time is dynamic rather than divisible into distinct segments or moments, it is experienced subjectively as duration. This subjective, inner time is less amenable to measurement because of its dynamic, subjective nature but, for Bergson, is clearly related to 'clock time' in that they work together, for example in the instilling of attitudes among workers such as 'time thrift' (Thompson, 1967), perceptions of the increase of the pace of life (Rosa, 2009)

and perpetual time (Wacjman, 2008). Psychologists have posited an innate sense of temporal rhythm that can be affected by factors such as emotional state (Benau & Atchley, 2020; Droit-Volet & Meck, 2007) and complexity of task requirements or multitasking (Kamp, Lund & Hvid, 2011). This subjective experience of time, or examination of the quality of time, is clearly important in understanding the time use of teachers. Our argument is that concerns about work intensification are concerns about 'lived time', or the subjective experience of certain periods of time. Beck (2017) uses the term 'heavy hours' to explain that subjective experience of work intensity as an individual feels themselves being pulled in multiple directions at once due to competing and contradictory demands. As Brante (2009) observes, time poverty explains the necessity of multi-tasking or synchronous work and the effect this has on levels of stress and wellbeing as the teacher has to 'triage' tasks. Multitasking can be understood as 'the simultaneous performance of several tasks, or the rapid alternation between them' (Spink et al., 2009 in Offer & Schneider, 2011). For example, experienced teachers will practice multitasking in the classroom, as they responsively move between teaching and classroom management. The experience of multitasking 'can create a greater sense of time, that is, it can deepen the intensity of time as well as maximise efficiency' (Offer & Schneider, 2011, p. 810).

Getting inside the subjective experience of time: The Teachers' Time Use app

The app we report on in this article has been designed to exploit the potential of contemporary technology, with minimal time demands on teachers. Harrison et al. (2019) used paper time use diaries with a group of early childhood educators to identify and quantify the complexity and intensity of educators' work. While they found this tool useful to gather detailed records, they also noted limitations in the detail of data recorded between various participants and found that the quality of data were compromised by a highly labour-intensive process. Similarly, Catzitheochari and Mylona (2022) have observed that time use diaries are burdensome and negatively affect response rates and data quality. Instead, they argue that new technologies offer opportunities for different methods of time use data collection. Indeed, random time sampling or the Experience Sampling Method (ESM)—which relies heavily on the concept of the time diary study—has become possible with advances in technology (van Berkel et al., 2017).

ESM offers a sustained method for capturing an individual's data over time, and also that individual's reactions to and beliefs about those activities (Forgasz & Leder, 2006). According to Csikszentmihalyi (1997, p. 15) ESM can provide 'a virtual film strip of daily experiences and activities' in a way that is less intrusive and less resource intensive. Importantly, as van Berkel et al., (2017, p. 93:4) observes, 'ESM reduces reliance on a participant's ability to accurately reproduce earlier experiences, minimising cognitive bias.' This is because ESM works in response to a signal that prompts participants to answer a question set. This enables a reduction in the time gap in-between onset of and reflection on the studied phenomena (van Berkel et al., 2017). While ESM was first trialled with electronic pagers that

notified participants to complete a paper self-report form upon each incoming signal (Csikszentmihalyi, 1977), mobile devices have allowed the evolution of ESM, including through the deployment of notifications and real-time digital data collection (Moreno et al., 2012), as well as advanced question logic and rich media collection (van Berkel et al., 2017).

While there is a range of ESM software for mobile devices these allow little flexibility in customising study parameters (see van Berkel et al., 2017). For us, customisation was important. We wanted to design an ESM interface that would work alongside the nature of teachers' work and the demands of classrooms. For this reason, we worked with an app developer to create our own ESM app, 'Teachers' Time Use'. Using the 'checklist for researchers' developed from van Berkel et al.'s (2017) systematic literature review of the use of ESM, we considered:

1. *Notification schedule*—on the morning of data collection, at the time of time use period, at the end of the day, and every 30 min as a reminder terminating at 7.30 pm
2. *Device ownership*—participants to use their own smart device (phone or tablet) with the app working on both android and apple operating systems.
3. *Participant compensation*—none
4. *ESM question type*—three input surveys delivered by Qualtrics; a before school survey, a time use survey and an end-of-day survey. These were embedded with a 7-point Likert sliding scale. The end-of-day survey also included opportunities for participants to provide open-ended responses.
5. *Advanced question logic*—the input surveys embed logical constructs for participants in school leadership positions, allowing them more categories of time use.

The app allows teachers to record 30-min of time use, and across all participants, it cumulatively measures a typical working week between the hours of 8 am and 4 pm Monday to Friday. We did not seek to measure 'non-working times' such as the evenings and weekends due to the potential encroachment on personal time. Instead, we asked participants to estimate in the end-of-day survey the amount of work still to be completed before their next working day.

Methods

Creation and pilot of the beta app

The *Teacher Time Use* app was designed with a commercial app developer through an iterative development process. First, the research team consulted the literature on how teacher work has been categorised. This resulted in a systematic review (see Creagh, 2023), and the creation of a list of categories to record teacher time use. This list informed the development of four meta categories of time use and nested sub-categories (see Table 1) that teachers could select from when using the app. Also embedded within the app were demographic questions about the participant

Table 1 Categories of time use

Meta category of time use	Time use activities
Face-to-face teaching activities	Learning interactions Feedback on student work Minor learning disruptions Significant behavioural interruptions Managing resources
Preparation and admin tasks	Planning and preparing lessons Data entry related to teaching, assessment and reporting Marking, feedback and other tasks related to assessment and reporting
Student wellbeing responsibilities	Out of class learning conversations with students Communicating with parents/carers about student learning Homeroom/pastoral care duties
Activities outside the classroom	Playground duties or supervisory roles Co/extracurricular activities Mentoring other teachers/supervising student teachers Additional/specialist roles Union duties Professional development All other meetings Email Other administrative duties

and their school, a before school survey to characterise stress at the beginning of the day, and an after school survey to understand time pressures across the day and into the evening or weekend.

Second, a pilot study was conducted in March 2022 with a small sample of teachers working across teaching areas and year levels ($N=8$). The purpose of this was to ensure the app's appropriateness and ease of use for recording time use activities. After the teachers provided informed consent, they downloaded the app, and utilised it to record 30 min of time use during a period of face-to-face teaching. Members of the research team observed participants during this 30-min period and made their own notes about time use for comparison and discussion in a follow-up interview with each teacher. The early feedback on the app was positive, with participants commenting that the app was easy to use even when they were busy; it did not significantly add to their workload; and the questions made sense and were easy to answer. Technical issues identified were addressed prior to the second pilot study. This small pilot was crucial in revising the app to ensure that it was fit for purpose.

Structure of the revised app

The revised app consisted of two stages: 'set up' and 'data collection'. Participants first downloaded the app, authenticating their participation with their system-provided email address. This email address was not recorded, instead a unique participant identifier was generated to enable linking of all responses for each participant. In the set-up stage participants answered demographic questions about themselves

and their school, and nominated three working days over a two-week period where they would be willing to record 30 min of their time use. The app then randomly allocated each participant a specific 30-min time slot for each of their three nominated days between 8 am and 4 pm. From here, the app moved to the second stage of data collection.

During data collection, the app sent notifications and reminders to participants to improve data collection. Participants received notifications on their mobile devices for each nominated data collection day. Notifications were sent every 30 min until completion. The surveys had to be completed in order of: 'Before School', '30 min Time Use', then 'After School'. Notifications stopped at 7.30 pm each night when participants were given the option to 'opt-out' of that day to allow for unexpected events. If a participant was unable to complete one of their nominated days and decided to opt-out, they could continue with their next designated day.

Participants

Participants for the pilot were recruited from the Queensland Teachers' Unions (QTU) Local Area Councils (LAC). LACs are elected, geographically grouped QTU committees located throughout the state. This convenience sampling meant that (a) there was a proportionally higher representation of participants working in regional schools in our sample and (b) the commitment of these participants to represent the QTU may suggest alignment with particular views regarding work. Given the pilot nature of this study, it was felt that the location and commitments of these participants were appropriate.

138 participants were recruited in October and November 2022 in Townsville, Mackay, Brisbane, Toowoomba, Sunshine Coast and the Gold Coast. Participants came from a range of ages and levels of experience, and mostly included teachers and some school leaders. Participants' schools were mostly regional and of average or lower than average ICSEA.[1] Primary schools and secondary schools were most commonly and evenly represented. Table 2 presents the demographic detail of the participant group and the schools they represented.

The analysis reported in this paper used data directly entered by participants into the app and embedded data fields generated by the survey software (Qualtrics) including times and dates for each survey, duration for completion of each survey, and identification codes for responses and participants. In the analysis that follows, we have prioritised findings which firstly indicate the utility of the app for time-poor teachers, and secondly which provide evidence of the subjective experience of work intensification, and for this purpose draw substantially on the qualitative data from the app. All statistical analysis was completed using Stata, Version 15.1.

[1] ICSEA stands for the Index of Community Socio-Educational Advantage. It is the common measure used in Australia to compare the educational advantage of a given school's student population.

Table 2 Demographic information of participants and the schools they represented ($n = 138$)

Demographic characteristics of participants	Number (rounded % of group)	Demographic characteristics of schools	Number (rounded % of group)
Age groups (years)		Type of school	
25 or less	6 (4)	Primary (P-6)	58 (42)
26–35	32 (23)	Secondary (7–12)	59 (43)
36–45	46 (33)	Primary and Secondary (P-10/12)	9 (7)
46–55	36 (26)	Special school/SEP (Prim & Sec)	9 (7)
More than 55	18 (13)	Other	3 (2)
Gender		Location of school	
Female	90 (65)	Major city	40 (29)
Male	47 (34)	Regional	79 (57)
Other	1 (1)	Remote	15 (11)
Years of experience			
0–5	18 (13)	Very remote	4 (3)
6–10	24 (17)		
11–15	30 (22)		
16–20	17 (12)		
More than 20	49 (36)		
Current position		Socio-economic status	
Teacher	40 (29)	Well above average	1 (1)
Senior teacher (ST)	13 (9)	Above average	12 (9)
Experienced ST	55 (40)	Average	46 (33)
Highly accompl. ST	1 (1)	Below average	54 (39)
HOD	9 (7)	Well below average	25 (18)
HOC	1 (1)		
HOSES	3 (2)		
Guidance officer	1 (1)		
Deputy principal	7 (5)		
Principal	7 (5)		
Head of school	1 (1)		

Results

Completion rates, times and coverage of the working week

Time use data were retained for all teachers who completed at a minimum the before school and thirty-minute time use survey, for at least one of their three nominated days. Of the 138 teachers who signed up for the app, 109 (79%) completed at least 1 day of surveys. Three timeslots were completed by 82 (75% of 109) respondents, 7 (6%) completed two timeslots, and 20 (18%) completed one timeslot. Participants generated a total of 815 surveys: 280 before school surveys, 280 30-min time use surveys, and 255 after school surveys.

Table 3 Completion rates for the time use surveys

Time Use survey	Mean (sd) in seconds	Median in seconds
Before school	132 (1474)	19.5
30 min of time use	293 (1534)	95
After school	170 (708)	61

The time use survey targeted a 30-min time slot, randomly allocated to teachers on each of the 3 days they selected when they set-up the app. Drawing on the concept of random time sampling (Bittman, 2016), this study was designed to build a dataset of the working week, compiling the combined experiences of multiple respondents, without overly burdening the respondents themselves. Random time allocation ensured coverage of the working day in 30-min time slots, from 8am through to 4 pm, Monday to Friday. For the pilot, even with a relatively small number of respondents, all time slots were covered for Monday through to Wednesday, with the exception of one 30-min timeslot on Thursday and three on Friday.

One of our concerns was the potential impost of completing the surveys for already time-poor teachers. However, none of the surveys exceeded an average time of five minutes for completion, and median times (in seconds) suggest completion was much quicker than five minutes for most participants. Table 3 provides mean, standard deviation and median time, in seconds, taken to complete each of the time use surveys.

We were interested in the time between the 30-min time slot allocated to teachers and their completion of data entry. To calculate this, we generated a time variable using the embedded data fields in Qualtrics. The results show the quickest time for submission at the end of the 30-min time slot was 55 s. 25% of the participants submitted their 30-min survey within 30 min of the end of the timeslot. The median was just over 2 h for submission, and the longest time for submission was 20 h, 6 min, 22 s. This outlier impacted the mean time, which was 7 h, 18 min and 45 s. With the obvious exception of the outlier, the median response of just over 2 h between designated 30-min time slot and data entry represents an obvious improvement of data collection in comparison to more traditional retrospective survey approaches. We note that this is a small, motivated participant sample and that patterns across a larger sample could vary. However, this shows that the app methodology seems to offer improvements in timeliness.

Time Use Survey 1 Before school

The before school survey consists of four questions about a participant's attitude toward the upcoming working day. For example, participants were asked the extent to which they were looking forward to the day ahead, how prepared they felt for the upcoming day and how well they had slept. Each question used a seven-point Likert scale ranging from one (not at all) to seven (to a great extent). There were 280 responses, from 109 teachers. Average responses to each of the before school survey questions indicated that teachers tended towards feeling positive in their anticipation of the day ahead, with average scores of 4.4 (SD 1.6) when asked if they were

looking forward to the day ahead, 4.5 (1.6) in relation to their preparedness for the school day, and 4.7 (1.4) in response to their sense of positivity about the day ahead. Across each question, there was greater variation in responses between teachers, compared to the variation within their individual responses.

Time Use Survey 2: Thirty Minutes of Time Use

Teachers were asked to record 30 min of activities from four broad categories and associated sub-categories of activities, presented in detail in Table 1. Each of these subcategories used a sliding measure for minutes spent on that task. Consistent with our aims of exploring teachers' subjective experiences of time use and multitasking, we purposefully did not provide teachers with a tally of the minutes as they recorded their activities or limit the number of minutes that could be recorded. Within a 30-min timeslot, the average total time reported on activities was 63.28 min. In other words, teachers indicated that, on average, they were doing 63 min of work in a 30-min time slot. The median was 50 min. For 75% of the 30-min time slots more than 30 min of time use was reported. This data were further broken down, to interrogate whether familiarity with the app would see a more 'accurate' recording of 30 min. When respondents recorded their first 30 min survey the mean time was 65 min of activities (SD 55), with a median of 55 min. For the second 30 min survey, again, respondents recorded 65 min on average (SD 58), with a median of 50 min. For the third engagement with the survey, the average was 59 min (SD 44, median 46 min). While some of this discrepancy might be accounted for by participants' over-estimation, even with increasing familiarity with the app, and a small decline in the average, these results suggest that teachers were reporting a 'complex temporal patterning of experience' (Wacjman, 2014, p. 15) due to engaging in multiple tasks simultaneously.

To illustrate, we can draw on both this study and the pilot to examine more closely the work of teachers when they engage in face-to-face teaching. Consistent with our early observations in the initial pilot, teachers multitasked when managing their classrooms. We observed teachers engaging in learning interactions while simultaneously managing resources or providing feedback to students, and reporting these simultaneous activities would account for the recorded time. Our data in this study suggest this is the case. Face-to-face teaching activities were reported in 153 (just over half) of the timeslots, with 64 of these time slots dedicated only to the face-to-face teaching category. The average time for the 153 timeslots was 46 min, (48 min on average for the 64 face-to-face only timeslots) and this time was allocated to a combination of the various options given in the app. Table 4 presents a breakdown of the combinations of activities reported by teachers whilst engaging in face-to-face teaching activities, and for one third of these time slots, teachers indicated that they were simultaneously engaged with all the activities listed for this category. Again, we recognise that this could be regarded as overestimation by teachers, or an alternate reading could be that these data reflect the layering of tasks, or the multitasking which constitutes classroom work.

Table 4 Face-to-face teaching activities ($n = 153$)

Learning interactions	Giving feedback	Minor disruptions	Significant behaviour	Managing resources	Frequency (n)	Percent
–	–	–	–	✓	1	0.65
–	✓	✓	✓	✓	1	0.65
✓	–	–	–	–	2	1.31
✓	–	✓	–	–	3	1.96
✓	–	✓	–	✓	5	3.27
✓	–	✓	✓	✓	3	1.96
✓	✓	–	–	–	8	5.23
✓	✓	–	–	✓	7	4.58
✓	✓	–	✓	–	1	0.65
✓	✓	–	✓	✓	1	0.65
✓	✓	✓	–	–	11	7.19
✓	✓	✓	–	✓	45	29.41
✓	✓	✓	✓	–	12	7.84
✓	✓	✓	✓	✓	53	34.64

We also calculated the total number of time use subcategories selected by teachers, across all activities provided in the app. The average number of subcategories selected was five (and this was consistent at each of the three timepoints), and the median 4.5. This means that on average, teachers were reporting 63 min of work across five different activities within a single 30-min timeslot. Figure 1 provides detail on the range of the number of tasks selected by teachers, spanning the 280 timeslots.

Figure 1 provides rich insight into the breadth of tasks teachers undertake. There was no subcategory without a time allocation across all time points, highlighting that teachers are juggling a wide range of tasks on a daily and weekly basis to attend to the multifarious demands of working in a school. Table 5 reports the number of timeslots in which each of the subcategories appeared, and the average number of minutes spent on each subcategory of activity. As would be expected, many of the time slots recorded 'core' tasks of face-to-face teaching, and significant preparation and teaching administration activities across the week in order to be prepared for face-to-face teaching. In addition to teaching and learning related activities, teachers were also involved in student welfare responsibilities outside of class time during their working week. Beyond teaching activities and care for students, teachers reported participation in many additional activities outside the classroom.

Time Use Survey 3: After School

While the before school survey gauged the readiness of participants for the school day ahead, the after school survey was designed to determine how the day had impacted teachers in terms of the manageability of the work, the extent to which they had felt rushed, and the amount of work that remained to be done before the

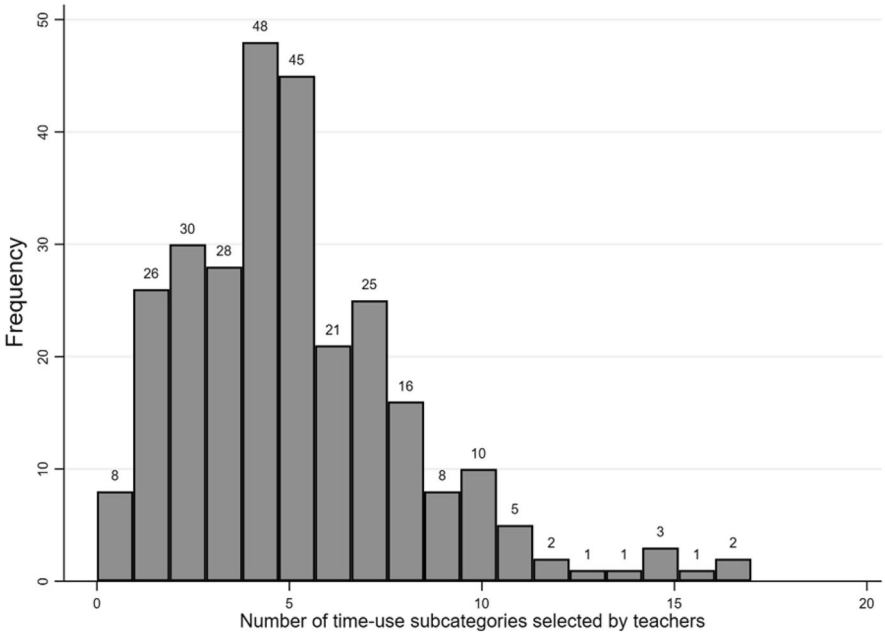

Fig. 1 Count of time use subcategories selected by teachers during 30-min time slot ($n = 280$)

next working day. We also wanted to measure the extent to which teachers felt that the day of the survey had been typical for them. The results of the after school survey suggest a more negative orientation on the part of teachers toward their work than that reflected in the before school surveys. In terms of manageability of workload for the day, the average response, on a seven-point Likert scale ranging from one (not at all) through to seven (to a great extent) was 3.6 (SD 1.5). Teachers were also asked to indicate, using the same seven-point scale, the degree to which their day had felt rushed. The average response was 4.8 (SD 1.6), indicating a tendency towards feeling rushed, rather than not rushed. For the after school survey, 100 teachers provided 255 responses, and again, the variability in response was greater between teachers, than within the clusters of their responses.

Participants were invited to provide further information about these two dimensions—of manageability or feeling rushed—in open-ended qualitative responses. 51 responses were recorded for manageability and 38 for feeling rushed. Coding of these two sets of responses suggests similarities in participants' reasoning across two dimensions. The top five reasons provided by participants for feeling that their work was manageable and rushed is summarised in Table 6, with examples. Interestingly, responses do not primarily concern face-to-face teaching; instead, all are to do with arguably 'non-core' aspects of teaching work, namely 'admin', extracurricular activities or curricular support beyond the classroom, working with other staff, undertaking leadership duties, and dealing with student welfare and behaviour issues. This suggests that while face-to-face teaching may constitute a large, if not the largest component of teachers' work in terms of clock time, this work does

Table 5 Number of time slots and average time for each subcategory of 30 min time use

Category and subcategories of activities	Number of timeslots where this activity was recorded ($n = 280$)	Average time spent on this activity (minutes)
Face-to-face teaching		
Learning interactions	151	21
Giving in class feedback on student work	139	11
Minor learning disruptions	133	9
Managing resources	116	8
Significant behavioural interruptions	71	7
Preparation and teaching admin		
Planning and preparing lessons	101	19
Data entry related to teaching, assessment, and reporting	51	13
Marking, feedback and other tasks related to assessment and reporting	46	13
Student welfare responsibilities outside class		
Out of class learning conversations with students	100	10
Homeroom/pastoral care duties	38	9
Talking to parents/guardians	36	8
Other activities outside the classroom		
Email	70	10
Playground duty/supervisory roles	68	16
Work related to additional/specialist roles	63	16
Administrative duties	48	13
Co/extra-curricular activities	41	14
All other meetings	37	16
Union duties	34	11
Mentoring other teachers/supervising student teachers	30	11
Professional development	16	16

not compress the subjective experience of time in the school day in the same way that more peripheral, additional or 'non-core' tasks can. These 'add-on' aspects of teaching work—which are to an extent unpredictable and sometimes unexpected (for more on this, see discussion of Table 7 below)—may therefore be important in explaining teachers' sense of feeling time-poor.

In the after school survey, we also sought an indication of the amount of work which remained to be done ahead of the next school day. Teachers reported between zero and 10 h of work, with a median of 2 h and a mean of 3 h of work remaining that evening or over the weekend. Figure 2 shows the range of hours of work remaining, reported by teachers across 255 days. For 50% of those 255 days, teachers reported between 1 and 4 h of work remaining to be done at home, and for a further 25% of those days, teachers reported between 4 and 8 h of work.

Table 6 Open-ended responses why respondents' workload did not feel manageable that day, and/or felt rushed

Theme	Reasons work did not feel manageable		Reasons work felt rushed	
	Frequency	Example	Frequency	Example
Admin	18	'Preparing and organising planning sessions and times for teams to develop 2023 curriculum teaching and learning plans, creating and modifying the 2023 GTMJs and tasks for all Prep to Year 6 KLAs made up a majority of my day'	20	'Too many emails'
Extracurricular activities or curricular support duties beyond the classroom	10	'I supervise basketball—takes my whole break time'	11	'Helping students prepare speeches for a mural dedication and helping make calls to prepare parents and students for camp next week'
Other staff	8	'Loss of senior leadership team members to illness meant a hard day'	7	'Dealing with supply teachers not attending'
Leadership duties	7	'Learning new processes as part of newly appointed promotional position'	1	'Tasks related to my HOD role'
Student welfare/behaviour	3	'Social Emotional support of students with needs outside lesson time'	9	'My day was chaos as 1 student required more support as his teacher was away. then i had to support a student that isn't officially on my case load'

We asked teachers to indicate whether the day of the survey was a typical working day for them, using a seven-point scale from 'not at all typical' to 'very typical'. The median for this question was five, indicating the majority of teachers had described a typical working day. Figure 3 shows the distribution of responses to this question.

The after school survey was useful in capturing a global view of a typical workday for the majority of respondents. Overall, the data suggest that managing the working day is a challenge for many teachers and feelings of being rushed are common. Alongside this, teachers anticipated an average of 3 h of work still to be done before the working day is finished.

Open responses from the final survey question asking for further commentary as to respondents' workload that day reveal further dimensions to respondents' reports that their work and workload was unmanageable, rushed and incomplete. The commonly occurring themes derived from a total of 84 responses, along with examples for each, are summarised in Table 7.

As with the previous open-ended responses, comments most frequently focused on aspects of work outside of face-to-face teaching. Key themes that emerged were the lack of consistency across the workday, the necessity to always be able to respond to change, and to think quickly to meet these challenges and disruptions. Importantly, this need to 'triage' time had affective impacts on teachers and their sense of how well, or successful, their days had been. Indeed, supporting the above analysis that it is 'non-core' work which contributes to feelings of time poverty, the two most dominant themes in this dataset were 'the unusual' and 'the unexpected'. It is further pertinent to note that although these were common themes, most participants also reported that their days were fairly typical (Fig. 3). This suggests that 'the unusual' is, in fact, usual in teaching; and given these varied, unpredictable, non-core aspects of the job are linked with feelings of work being unmanageable and rushed, the typicality of 'the unusual' in teaching may be a key factor in widespread experiences of time poverty.

Discussion

Understanding the granularity of teachers' work is important to form coherent and incisive policy responses to teachers' work. However, this understanding is hampered by methodological difficulty. Measuring teachers' work is simple if the aim is to understand the 'gist', but difficult if the aim is more exact and specific information (Braak et al, 2022). Further, an ethical challenge for researching teachers' work concerns what is reasonable to ask of already overworked teachers in order to better understand, and potentially respond to, that overwork? With that in mind, the Teachers' Time Use app was designed to enable collection of more granular data without creating too much of a time impost on teachers and school leaders.

The pilot study suggests that using an ESM (Larson & Csikszentmihalyi, 2014) approach to measuring teachers' time use has merit. First, attrition was relatively low for a data collection instrument that asks participants to input data at nine points on three working days over a 2 week period. The app was downloaded and

Table 7 Comments about workload that day

Theme	Frequency	Example
The unusual	37	'Was harder than usual after mandated awards night attendance previous night to 8:30 pm'
The unexpected	16	'Term 4 is crazy & finding out I have move classrooms has added an extra layer of stress'
The juggle	13	'Every individual job is very doable, the issue is the competition of priorities and the lack of time'
The affective	8	'It was a struggle and a very hard day'
The personal	4	'Problems with my partners roster/job attributed to stress and not completing jobs I wanted to due to family responsibilities'
The admin	4	'OneSchool contacts took up a lot of time today'

demographic surveys were completed by 138 teachers, and 79% of this group (109 teachers) completed time use surveys for at least 1 day. While this may partly be explained by the motivation of an atypical sample, poor design or an excessive time impost would extinguish that motivation. Second, it enabled the random allocation of 30-min slots of time use. While participants chose the days that they would record their time use, the app allocated each 30-min time slot for each participant. This random allocation improves the reliability of the data gathered in that it removes the possibility for participants to choose their most extreme class or period. Third, one of the challenges with recording teachers' time use concerns the limits of memory on recollection. Asking participants to recall time use over the last week, or last month, or last year introduces significant uncertainty into the data collected. This does not necessarily make such recollections wrong, however recalling more precise detail becomes less easy over time (Brainerd & Reyna, 1993). Reducing the time between an event and recall of that event can improve the accuracy and detail. As the results show, 50% of the data submissions for the 30-min time slot were done within 2 h, and a further 25% were submitted in the range of 2 h to 6 h after the time slot. This does not completely solve the problem of relying on participant memory, however, it improves the accuracy of each recollection.

This approach to collecting data on teachers' work has the potential to change how we understand that work and where problems reside. Currently, our conversations are largely focused on workload, and policy solutions are designed to address the problem of too much work. However, as the research literature suggests, workload as the hours spent on work over a given period is only one aspect of teachers' work and related concepts such as heavy hours are equally important (Beck, 2017). Essentially, it is not just how much work teachers have to do, it is also how complex, demanding, difficult and/or stressful their job is at given moments that must be understood. The Teachers' Time Use app gets inside the hours that teachers work and provides insight into what makes some hours heavier, or more demanding, than others. This offers critical insight into teachers' work and suggests new possibilities of intervention when that work has become unmanageable or unbearable.

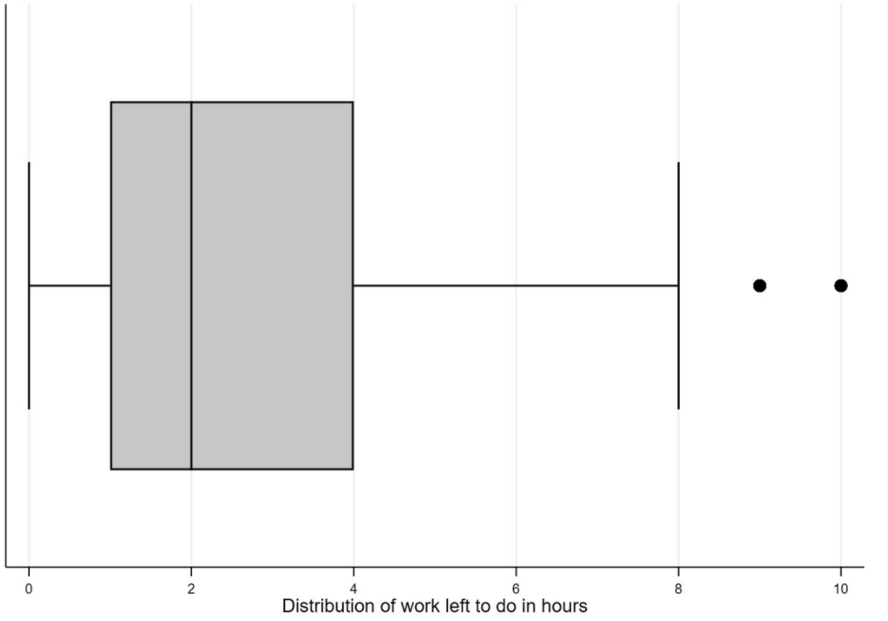

Fig. 2 Hours of work remaining to do ($n = 255$)

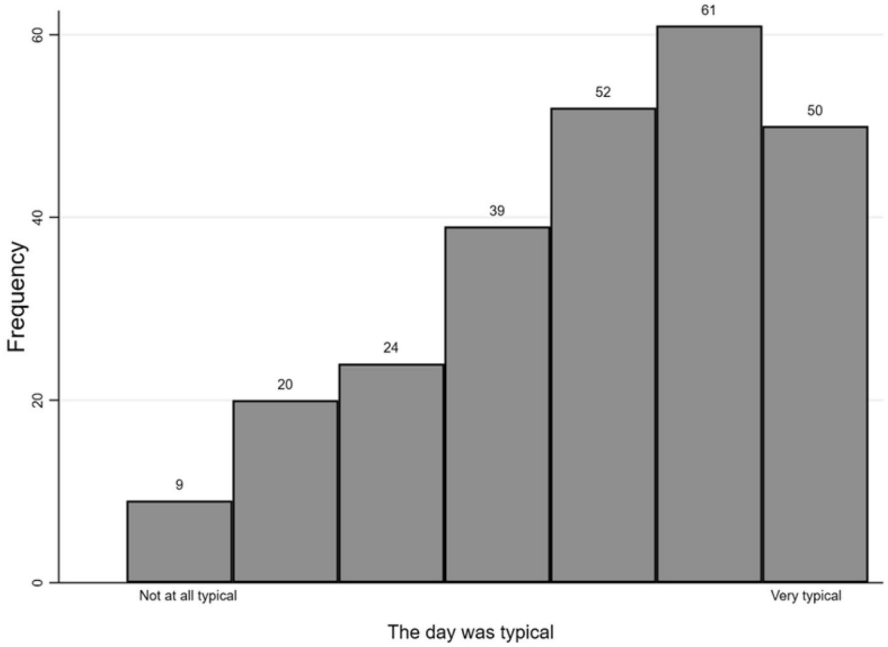

Fig. 3 Extent to which the day was typical ($n = 255$)

One of the more significant, yet difficult, findings concerns the allocation of time to tasks across the 30 min segments. The allocation of an average of 63.28 min of time use to specific tasks across 280 recorded 30-min time periods appears counter-intuitive. As we have outlined previously, our aim in the project is to try to understand the *subjective* experience that teachers have of time as a way of understanding work intensity and/or intensification. For this reason, when we designed the app we made a conscious decision to allow open, or free, estimation of time use across the four categories and sub-categories. This means that rather than forcing the time use recorded to equal 30 min, participants were able to allocate more than 30 min to tasks. There were a number of reasons for this: first, the literature that addressed the experience of time, and feelings of being time-poor, or that the pace of work was accelerating, etc. stress that these are *subjective* experiences (Rosa, 2009; Wacjman, 2008; Wacjman, 2014). Second, there was a concern that if participants could only allocate 30 min of time use, they would fail to record all aspects of their work. Finally, and related to the previous points, significant research points to the craft of teaching as developing skill in rapid switching between, and layering of, tasks as teachers manage pedagogic, curriculum, lesson sequencing, behaviour, interpersonal relations and so on (Berliner, 2004). However, the scale and scope of switching and/or multitasking; the cognitive demands associated with switching domains; and how problem solving and decision making across these multiple domains impacts individual teachers are not well understood. Allowing open estimation of time use was the best way to get inside this subjective experience of time, to understand why it appears intense and, in future, for whom. In measuring subjective experience it could be claimed that we are introducing some imprecision into the findings, but this is defensible both logically and theoretically. Teachers are not leaving the profession because they find the notion of organising time into hours and minutes problematic, they are leaving because of their subjective experience of what is happening within those time intervals.

The complexity of teachers' subjective experience of work appears further impacted by the amount of work that teachers felt they needed to complete after hours. On average, teachers reported that they still had 3 h of work left to do that night or over the weekend. Thus, teachers using the app are reporting both the heavy hours of their work (intensity) and the amount of work they have to do (workload)—and, arguably, the interaction of these forces, resulting in a 'residue' (Beck, 2017) of work to be completed at the end of the day. This is shedding light on what Wacjman (2014) posited as an increasingly common experience of modern life, that of feeling 'time poor'. Our argument is that this time poverty is an effect of the amount of work to be done and the subjective perception of the intensity of that work.

In general, it seemed that the participants had a relatively positive attitude when starting the school day. Responses indicating how positive participants felt about the upcoming day and how prepared they felt for the school day tended toward the more optimistic end of the scale. However, at the end of the day the questions about how manageable their workload was that day and how rushed they felt during the day tended toward a more negative overview of the day. Those who recorded 'higher'

levels of dissatisfaction with their workload and who felt rushed were asked to qualify factors that impacted this experience. Participants identified three common factors in response. These were managing student needs/behaviour, communicating with parents/carers and the amount of work to be covered in lessons. It is easy to see how these factors add layers of complexity to teaching. Managing student needs and/or behaviour can take time away from teaching and learning activities, time which is at a premium where teachers feel pressure to keep up with syllabus and curricular content. Communication with parents, whether via email or through a student management system, similarly becomes an administrative task that must be done on top of an already intense schedule (Heffernan et al., 2022).

Participants were also able to indicate other factors that impacted their workload and feelings of being rushed. The overwhelming theme arising was 'the unusual'. For instance, one respondent described their day as 'harder than usual after mandated awards night attendance' the previous night; another commented that they had 'actively supervised students from 4:00 to 7:30 pm due to school disco night'; others said that there had been 'a staff briefing during lunch' or a '1 h meeting before school'. This overriding theme of 'the unusual' was, somewhat paradoxically, so common that it could be argued that the unusual is usual in schools; that day-to-day routines feature so much interruption that interruption becomes a norm. This is further supported by the second most common theme in the data, which we describe as 'the unexpected'. For while many of the 'unusual' activities described by respondents may have been planned, others were not. Respondents described, for example, having a 'usual full on work day on top of dealing with behaviour of other students NOT on my support workload', or that 'finding out [they] have [moved] classrooms has added an extra layer of stress' to the day. 'Too many disruptions' and 'general disruptions' were similar general points of commentary, with a further two respondents also describing being assigned unexpected playground duties which impacted preparation time. For other respondents, meanwhile, it was the particular combination of activities: that it was the 'massive juggle' of their day overall, rather than any particular task, that they wished to highlight. Indeed, a further theme arising reflected the affective impact of the intensity and load of teaching, with respondents commenting for example: 'teachers' days are exhausting', 'it was a hard day', 'it was a struggle and a very hard day'.

Perhaps the final point of interest concerns the range of responses. While on average teachers reported significant concerns with their work, there remain a minority within this group for whom workload and work intensification, or time poverty, were less of a concern at the time of data collection. 11% (28 out of 255) of the entered data in the after school survey reported that for that day, they had less than 1 h of work to do that night, for 22% (56 out of 255) end-of-day surveys, there was little or no experience of feeling rushed and 26% (67 out of 255) entries reported that they found their day very or relatively manageable. A larger sample would allow these patterns to be better understood, but it points to the variety and complexity of teachers' work.

Conclusion

The Teachers' Time Use app responds to the challenge of better measuring time use without adding a significant burden to already overworked teachers and school leaders. This pilot suggests that the app can 'get inside' the intensity of teachers' work to explore what makes some hours heavier, or more demanding, than others. In this, it addresses the experience of time poverty, understood as the intersection of workload and work intensification. It shows promise in advancing understanding, and potential systemic solutions, to the problems of teacher time use that are being referred to as an impending crisis. It adds to what research knows about workload, and sheds light on the layering of activities, tasks and disruptions, and the cost of those responses, that teachers are routinely asked to manage.

As a pilot there are limitations. While there was diversity of participants and school characteristics, the sample is small and atypical. Results should not be presented as generalizable to teachers across Queensland. The pilot shows that the app is a feasible means of recording teachers' time, and therefore it would be worthwhile recruiting a larger sample. A larger sample would also add robustness for more detailed multilevel statistical analysis of subgroups to better understand the characteristics of those most likely to be experiencing time poverty, and in what contexts this is most likely to occur. This is a critical first step in thinking through what might be done by education systems to better balance teachers' work demands.

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Declarations

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